

Bijal Panchal

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SUMMARY

Data Analytics Professional (M.S. Data Analytics, April 2026, Graduated with Distinction — GPA 3.89/4.0) specializing in ETL pipeline development, data warehousing, and business intelligence. Strong track record of delivering production-ready solutions: designing star schema warehouses consolidating millions of records, building automated Python pipelines that reduced data inconsistencies by 30%, and developing 10+ Tableau and Power BI dashboards directly adopted for business decision-making. Combines analytical rigour with machine learning expertise — including time series forecasting, RFM segmentation, and customer segmentation — to transform complex datasets into measurable business outcomes. Proficient in Python, SQL, Power BI, Tableau, Pandas, Scikit-learn, MySQL, and SQLite.

SKILLS

Programming & Libraries — Python, SQL, Pandas, NumPy, Scikit-learn, Statsmodels, XGBoost, Prophet, Matplotlib, Seaborn, SQLAlchemy, SciPy, Requests, BeautifulSoup4

Data Visualization & BI — Power BI (DAX, Power Query), Tableau, Microsoft Excel (Advanced), Word, PowerPoint

Machine Learning & Analytics — K-Means Clustering, XGBoost Classification, PCA, ARIMA, SARIMA, Holt-Winters, Isolation Forest, Z-Score Anomaly Detection, RFM Segmentation, Time Series Forecasting, Feature Engineering

Databases & Data Engineering — MySQL, SQLite, PostgreSQL, Star Schema Design, ETL Pipelines, Data Warehousing, Data Modelling

Tools & Platforms — Jupyter Notebook, VS Code, Git, GitHub, DBeaver, Streamlit, GitHub Actions, pytest

EXPERIENCE

POINTS AFRICA

Toronto, ON(Remote)

Data Analyst Intern

Sep 2025 – Jan 2026

- Processed and standardised datasets from 3 partners (GLICO, JUMIA, STAR) across core data domains including customer profiles, transactions, loyalty ledgers, and geographic data using Python (pandas, NumPy) and SQL, reducing data inconsistencies by 30% through automated cleaning pipelines.
- Enforced 100% transactional field coverage through completeness, accuracy, and consistency checks — resolving phone formatting errors, date-time mismatches, and duplicate records via iterative Python ETL workflows.
- Architected a star schema data model with 4-dimension tables (Customer, Partner, Store, Date) and a central Transaction Fact table, consolidating all partner data into a unified master dataset and improving cross-partner query performance by 40%.
- Applied RFM segmentation modelling to classify customer loyalty tiers, identifying high-value clusters across 3 transaction networks and enabling targeted partner engagement strategies.
- Designed and delivered 10+ production-ready Tableau dashboards covering KPIs, partner performance, payment behaviour, and geo-regional analytics, directly adopted for internal business decision-making.

DMG GROUP OF COMPANIES

Ahmedabad, IND

Data Analytics & ML Intern

Jun 2024 – Nov 2024

- Designed and implemented a customer segmentation model for IT service clients using Python (Scikit-learn, pandas), applying K-Means Clustering and XGBoost classification, achieving 87% classification accuracy in identifying high-value client segments.
- Engineered 12+ behavioural and transactional features from client service usage and contract data, reducing data noise by 28% through advanced feature selection and dimensionality reduction (PCA).
- Built interactive client performance and revenue dashboards in Excel and Power BI, consolidating data across 4+ service verticals, enabling real-time tracking of client acquisition, service utilization, and revenue trends.
- Performed data cleaning and ETL workflows on 4,000+ client records using Python and MySQL, standardizing inconsistencies across service and billing datasets, improving overall data quality by 30%.

- Documented model evaluation metrics (Precision, Recall, F1-Score, AUC-ROC) and presented actionable insights to management, directly supporting data-driven decisions that improved client retention by 18%.

Projects

Global City Traffic Intelligence Platform | Python, SQLite, Streamlit, Scikit-learn, Prophet — Live

- Engineered a fully automated real-time data platform collecting daily metrics across 5 global cities from 5 APIs (TomTom, OpenAQ, Open-Meteo, Numbeo, City Open Data), building modular Python collectors into a SQLite star schema warehouse (9 tables, 4,400+ rows) with a GitHub Actions daily pipeline running at 06:00 UTC.
- Designed a weighted City Stress Index scoring engine (35% Traffic + 25% Air Quality + 20% Weather + 10% Safety + 10% Cost of Living) with 15+ engineered features including rolling 7/30-day averages, volatility scores, and heat stress index, powered by ARIMA, Prophet, Z-Score, Isolation Forest, and K-Means analytics models.
- Deployed a 7-page interactive Streamlit dashboard (city-stress-index.streamlit.app) featuring live stress score cards, 30-day forecasts, anomaly detection alerts, K-Means city clustering, and cross-city radar comparisons, backed by a pytest suite of 22 tests covering database integrity and pipeline logic.

Retail Sales Forecasting & Demand Planning | Python, SQL, Power BI, Statsmodels, SQLite

- Ingested and cleaned 1M+ transactional records across 43 countries and 4,735 products using Python (Pandas, NumPy), resolving 34K duplicates, 19K cancellations, and 243K missing customer IDs, then loaded into a SQL star schema warehouse (SQLite, 1 fact + 4 dimension tables) via a Python ETL pipeline (SQLAlchemy).
- Built and compared SARIMA and Holt-Winters (Statsmodels) forecasting models on monthly revenue data, with SARIMA achieving 74.5% accuracy (MAPE 25.54%) on a chronological 80/20 train-test split, reducing simulated stockout rates by 18%.
- Delivered a 4-page Power BI dashboard featuring dynamic KPI cards, seasonal trend lines, actual vs. forecast comparison with 80% confidence intervals, inventory risk flags, and RFM segmentation across 6 segments — identifying 843 Champions averaging £5,689 spend and 409 At-Risk customers as the top re-engagement priority.

EDUCATION

University of Niagara Falls

Master of Data Analytics

Niagara Falls, CA

Oct 2024 – Apr 2026

Navrachana University

Bachelor of Technology – Computer Science & Engineering

Vadodara, IND

Jul 2020 – Jun 2024

CERTIFICATIONS & ACTIVITIES

Advanced Power BI: DAX Language, Formulas, and Calculations | Course Completion – Apr 2025

- A targeted Power BI certification covering core skills in **data modeling**, **DAX language**, and **advanced DAX formulas**. Emphasizes practical application through **dynamic dashboards** and **real-world business intelligence scenarios**.

Hack For Change | Participant – Apr 2022

- Participated in a **49-hour hackathon** organized by Navrachana University in collaboration with NuvYuva.
- Collaborated with peers to develop **innovative solutions** aimed at **social impact** and **technology-driven change**.
- Strengthened skills in **project development**, **teamwork**, and **rapid problem-solving under time constraints**.